

Day 1 — The Hook — What Is the Trick?

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 23×11 , 35×11 mentally using the split-and-sum method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthaji”)

Primary source (today’s reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Compute 23×11 silently in your head.** Pause. Show of hands for who got 253. Then write $23 \times 11 = 253$ on the board.
2. **Reveal the trick.** Split the digits of 23 (gap in the middle): **2 _ 3**. Add the digits in the middle: $2 + 3 = 5$. Fill the gap: **2 5 3** = 253. Done.
3. **Try together.** $35 \times 11 = 3_ (3+5)_ 5 = 385$. $24 \times 11 = 264$. $71 \times 11 = 781$. $52 \times 11 = 572$. $80 \times 11 = 880$. Pace: 4 problems in 2 minutes.
4. **Tirthaji’s framing.** This method appears as an application of the sūtra *Ekādhikena Pūrveṇa* (“by one more than the previous”) in Bharati Krishna Tirthaji’s *Vedic Mathematics* (1965). Honest historical note: the manuscript was written 1911–1918; some scholars (Dani 1993) consider it a 20th-century mental-arithmetic system rather than strictly ancient mathematics.

5. **Independent practice (5 min).** Worksheet: 20 two-digit $\times$ 11 problems, no carry case. Pairs compete.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Hook — What Is the Trick?.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).